

Curriculum Vitae / Resume

PRANAV JHA

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Education

University of Michigan

Master of Science in Robotics

Aug 2026 – May 2028 (Expected)

Ann Arbor, MI

Manipal University Jaipur

BTech (Hons) Computer Science and Engineering (IoT and Intelligent Systems) CGPA-7.77 (3.32/4)

Sep 2022 – Jun 2026

Jaipur, RJ

Experience

• InterraIT

Student Trainee

Noida, Jan 2026 – Apr 2026

- Developed a **digital twin of the FANUC R-50i Robot Controller** using **NVIDIA Isaac Sim**, enabling realistic simulation, performance analysis, and virtual validation of robotic operations.
- Explored **ROS2 integration** with NVIDIA Isaac Sim for simulation-driven testing, control, and validation of robotic applications.

• CSIR – National Institute of Oceanography (CSIR-NIO)

Research Intern

Goa, June 2025 – Aug 2025

- Utilized **SegFormer**, **UNet**, and **PIDNet** architectures with **Knowledge Distillation (Teacher-Student Framework)** to develop lightweight, **real-time coral reef segmentation models** for underwater imagery.
- Conducted dataset preprocessing and augmentation on coral reef datasets to improve robustness under low-light and turbidity conditions.
- Deployed optimized models on NVIDIA Jetson-Nano** for live inference through an ROV (Remotely Operated Vehicle), achieving efficient underwater object segmentation and classification.

• Siemens Digital Industries Software

IoT Intern

Gurgaon, May 2024 – July 2024

- Developed and integrated **real-time analytics pipelines** on edge IoT devices using **Azure AI and Azure IoT Hub** to monitor industrial sensor data.
- Automated large-scale Excel-based workflows and data synchronization, reducing manual reporting time by **15%** and improving data consistency.
- Explored **PLC-IoT integration and mobile device management** for intelligent industrial automation under Siemens DISW mentorship.

• Indira Gandhi Delhi Technical University for Women (IGDTUW)

Deep Learning Intern

Delhi, June 2024 – Aug 2024

- Designed and trained a **Heart Disease Prediction Model** using advanced **feature engineering and ensemble learning**, achieving a 30% improvement in diagnostic accuracy.
- Built a **waste classification system** leveraging **convolutional neural networks (CNNs)** to enhance sorting efficiency by 20% for sustainable waste management.
- Conducted comparative analysis of models using TensorFlow and PyTorch to identify trade-offs between speed, accuracy, and computational cost.

• CodSoft

Machine Learning Intern

Remote, Jan 2024 – Feb 2024

- Built a **Movie Genre Classification Model** using **TF-IDF** and **Multinomial Naive Bayes** to predict movie genres from plot summaries via **NLP**.
- Developed a **Credit Card Fraud Detection** system using **Logistic Regression** to identify fraudulent transactions in real-world datasets.

- Created an **SMS Spam Classification** model using **TF-IDF** and **Multinomial Naive Bayes** for spam message detection.
- **Bharat Intern** Remote, Jan 2024 – Feb 2024
Machine Learning Intern
 - Designed and trained **machine learning models** for data-driven classification and prediction tasks using **Python, Scikit-learn, and Pandas**.
 - Implemented **end-to-end pipelines** for preprocessing, model training, and performance evaluation across multiple ML projects.

Projects

- **Digital Twin of FANUC R-50i Robot Controller** Jan 2026 – Apr202
NVIDIA Isaac Sim, ROS2, Digital Twin, Industrial Robotics
 - * Developed a **digital twin** of the **FANUC R-50i Robot Controller** using **NVIDIA Isaac Sim** for realistic simulation and virtual validation of industrial robotic operations.
 - * Integrated **ROS2** with Isaac Sim to enable simulation-based testing, controller validation, and performance analysis before real-world deployment.
- **Coral Reef Semantic Segmentation System** June 2025 – Sept 2025
SegFormer, UNet, PIDNet, Jetson Nano
 - * Designed a lightweight semantic segmentation system using teacher–student distillation.
 - * Deployed models on Jetson Nano for real-time underwater coral classification on ROV footage.
- **Autonomous Firefighting Robot** Jan 2025 – May 2025
ROS Noetic, Gazebo, RViz, YOLOv8, LiDAR, SLAM, SDF
 - * Designed and simulated a fully autonomous robot for detecting and extinguishing fires using SLAM-based navigation and thermal vision.
 - * Integrated YOLOv8 and thermal sensors for fire detection, and used a robotic arm to launch a fire extinguisher ball with kinematic accuracy.
 - * Built in SDF format with differential drive, camera, LiDAR, arm, and fire plugin in a Gazebo environment.
- **GestureSync: Gesture-Based Virtual Control System** Jan 2025 – Mar 2025
MediaPipe, PyAutoGUI, OpenCV
 - * Created a hands-free keyboard, mouse, and media controller using real-time gesture tracking.
 - * Enabled actions like play/pause, volume, and cursor movement via fingertip detection.
- **Real-Time Sign Language Detection System** Aug 2024 – Oct 2024
MediaPipe, TensorFlow, OpenCV
 - * Built a CNN-based model to recognize static and dynamic gestures using MediaPipe landmarks.
 - * Enabled real-time sign-to-text translation for hearing- and speech-impaired users.
- **P.P.E Detection Kit** Jan 2024
Arduino UNO, ESP32, Servo Motor, Python, C, YOLOv8
 - * Developed a smart safety gate that verifies PPE compliance: helmet, mask, safety goggles, and vest.
 - * Integrated YOLOv8 detection with Arduino-controlled servo motor to grant access only upon full gear compliance.
- **Smart Attendance System** Oct 2023
ESP32, RFID, Arduino IDE
 - * Designed a custom PCB device for attendance automation using RFID-based ID card scanning.
 - * Logged entries and exits at doors via ESP32-controlled system integrated with entry tracking logic.

Technical Skills

Languages: Python, Java, C, HTML, CSS, JavaScript, PHP*, SQL

Developer Tools: VS Code, Eclipse, Arduino IDE, PyCharm, Mendix

Technologies/Frameworks: Linux, GitHub, Computer Vision, Machine Learning, IoT, Arduino, Raspberry Pi, Nvidia Jetson-Nano, Deep Learning, Node-RED*, ROS, Generative AI, 8051 *Elementary proficiency

Certifications

Cisco: CCNA – Introduction to Networks & Introduction to Switching, Wireless, and Routing Essentials
University of California, Irvine: Introduction to IoT (6-course specialization)
Microsoft: Designing and Implementing Azure Solutions on Edge Devices
Stanford University: Supervised Machine Learning: Regression and Classification

Patents, Publications and Other

• Publications:

- * *Knowledge Distillation of Real-Time Coral Segmentation for Underwater ROV Applications* — Ongoing Journal Publication (Nov 2025)(2nd author)
- * *Real-Time Coral Semantic Segmentation for Underwater ROV Applications* — Ongoing conference publication waiting to be accepted (INDIAcom 2026) (Dec 2025)(2nd author)
- * *Autonomous Firefighting Robot with Catapult-based Suppression in ROS-Gazebo Environment* — Accepted and presented at Conference, awaiting Publication (Nov 2025) HITA 2025(1st author)
- * *Leveraging Optimized Hybrid Ensemble Models to Predict Sustainability Awareness in Indian Youth* — Presented at Conference COACONS 2025,awaiting Publication (Mar 2025)(2nd author)
- * *Blockchain-based Optimization Algorithm for Secure IoT Communication Using AMQP* — Published in IEEE Xplore (Jan 2025) ICAIHC 2025(3rd author)

• Patents:

- * **An IoT-based Smart Parking System using Arduino UNO and Xcode** Patent No: 202511067693
- * **A Fire-fighting and rescue robot using real time image detection and autonomous navigation in indoor environments** Patent No: 202511066903
- * **GestureSync Music Controller** Patent No: 202411086632
- * **People Counting Device** Patent No: 202311060355
- * **Smart Attendance System** Patent No: 202311060429
- * **Automated Fare Collection Device** Patent No: 202311060360

Positions of Responsibility

• Technical Secretary, IEEE Sensors Council MUJ (June 2024 – April 2025)

Oversaw technical documentation and communication within the council. Assisted in organizing workshops and events promoting sensor innovation. Facilitated inter-team collaboration aligned with council goals.

• Head of Projects, IEEE Sensors Council MUJ (May 2023 – June 2024)

Led project planning and execution. Organized flagship events including:

- * **Talkathon - INNOVATEX** featuring Mr. Sanchit Jain (Founder, Knowledge Gate)
- * **Talkathon - INNOVATEX 2.0** featuring Mr. Varun Singla (Founder, Gate Smashers)
- * **Time-Bat Relic:** An online treasure hunt event

Workshops

• Guest Speaker – Introduction to Robotics Workshop at School (May 2026)

Invited as a guest speaker to conduct a robotics workshop for students of Classes 6–10. Delivered an interactive session on modern robotics, emerging technologies, and career opportunities, followed by a hands-on laboratory session where students built and explored two robotics projects.

• Introduction to Robotics Workshop at School (Aug 2024)

Conducted a hands-on robotics workshop as an invited alumni for students of Classes 6–8. Introduced fundamental robotics concepts and real-world applications. Demonstrated basic robotics projects and discussed future prospects in robotics across industries.

Achievements

- Selected among **Top 200 teams nationwide** in the **Smart India Hackathon (SIH)** representing college (2024).
- Awarded a **Certificate of Appreciation** by the Head of Department for successful patent publication (2023).
- Awarded a **Certificate of Excellence** by the college for Outstanding performance in the field of **Technical and Innovation**(2025).

- Awarded a **Certificate of Excellence** by the college for Outstanding performance in the field of **Research** (2025).
- Awarded another **Certificate of Excellence** by the college for Outstanding performance in the field of **Research** (2025)